Experiment - 11

Routing at Network Layer

Aim :

To Stimulate Static Routing and RIP using Cisco Packet Tracer

A) Static Routing :

Static routing is method where routes are manually added to the router routing table. It is useful for controlling path taken by Packet bitt network.

Procedure :

1) open cisco packet tracer
2) Place required routers, Switch and Pc in workspace
3) connect the devices using suitable cables
4) assign IP address to all PCs and router interfaces
5) open the router CLI
6) Enter the foll command to configure static route
7) Use foll command to check routing Show IP route

B) RIP Routing :

RIP is a dynamic routing Protocol. It allows routers to automatically exchange information abt the network connected to them. RIP Select routes based mainly on

Procedure :

1) Create network with three routers and PCs shown in Packet tracker topology

2) Assign IP address to PCs and router interfaces

3) Configure the route interface using

4) Repeat RIP configuration for other router with their directly connected

5) Repeat RIP configured for other router with their directly connected network

6) Allow some time for routers to exchange routing information

7) Send simple PDV from source PC to destination PC and observe Packet Path-

Observation:

In Static routing, router are manually configured by administor

In RIP, router automatically exchange routing information

RIP selects a route based on lowest hop count

Result:

Thus, Static routing and RIP were successfully simulated using Cisco Packet Tracer